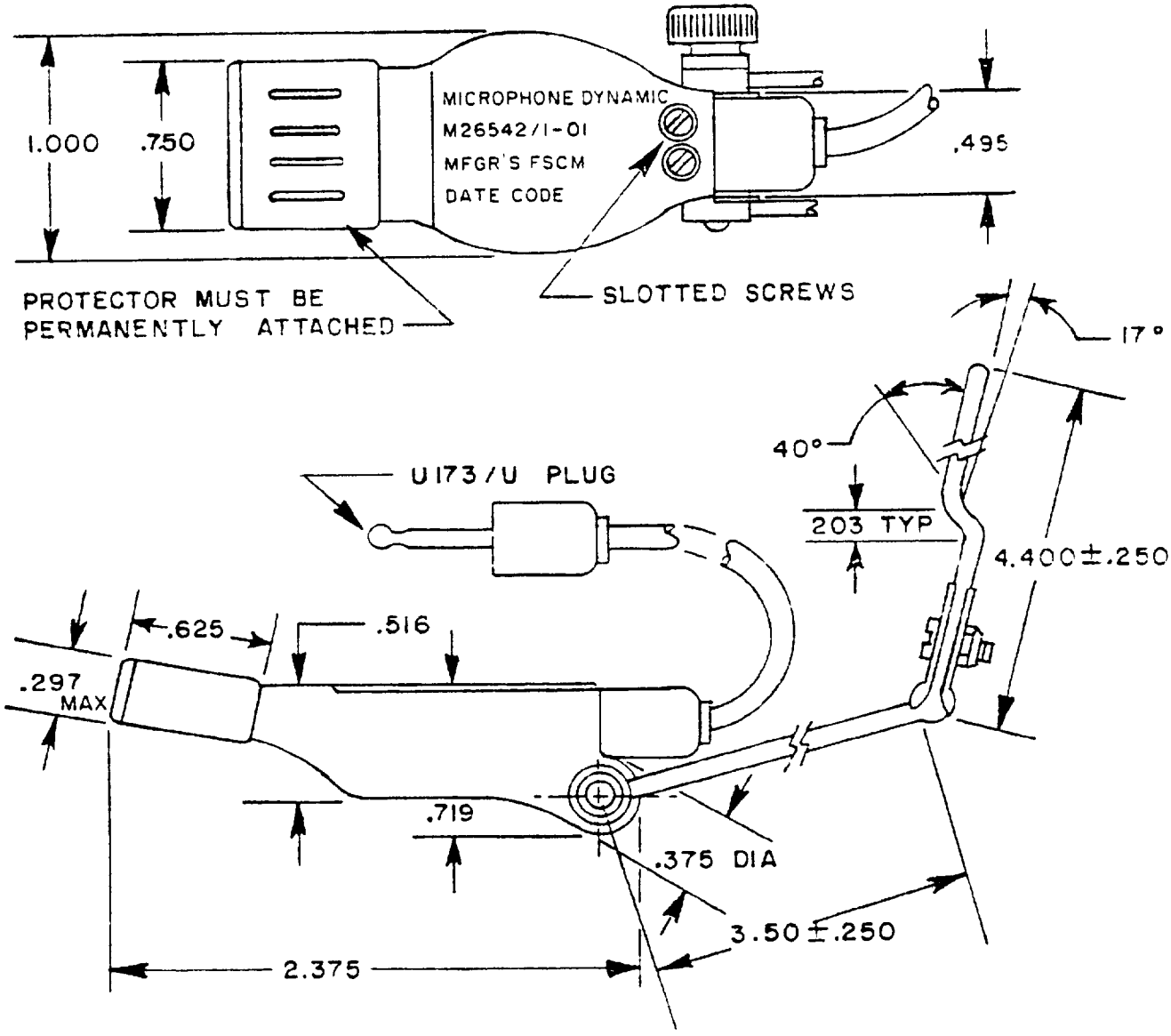
MILITARY SPECIFICATION SHEET

MICROPHONE, M26542/1-01

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

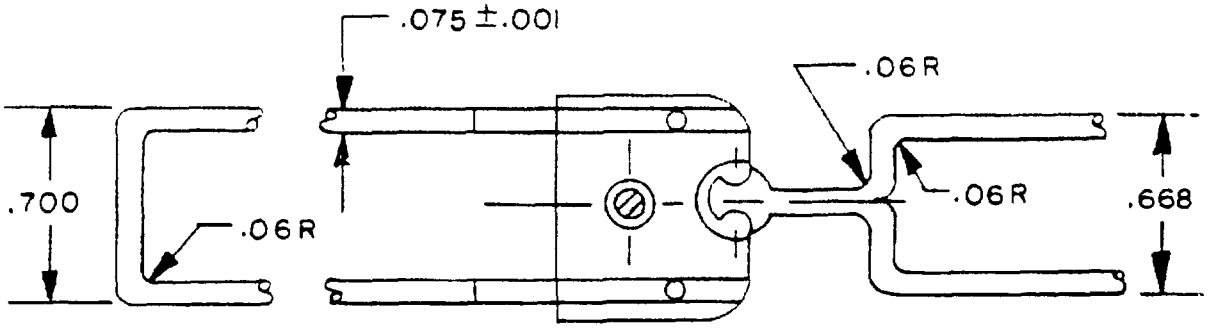
The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.



© FIGURE 1 Microphone assembly

© denotes changes

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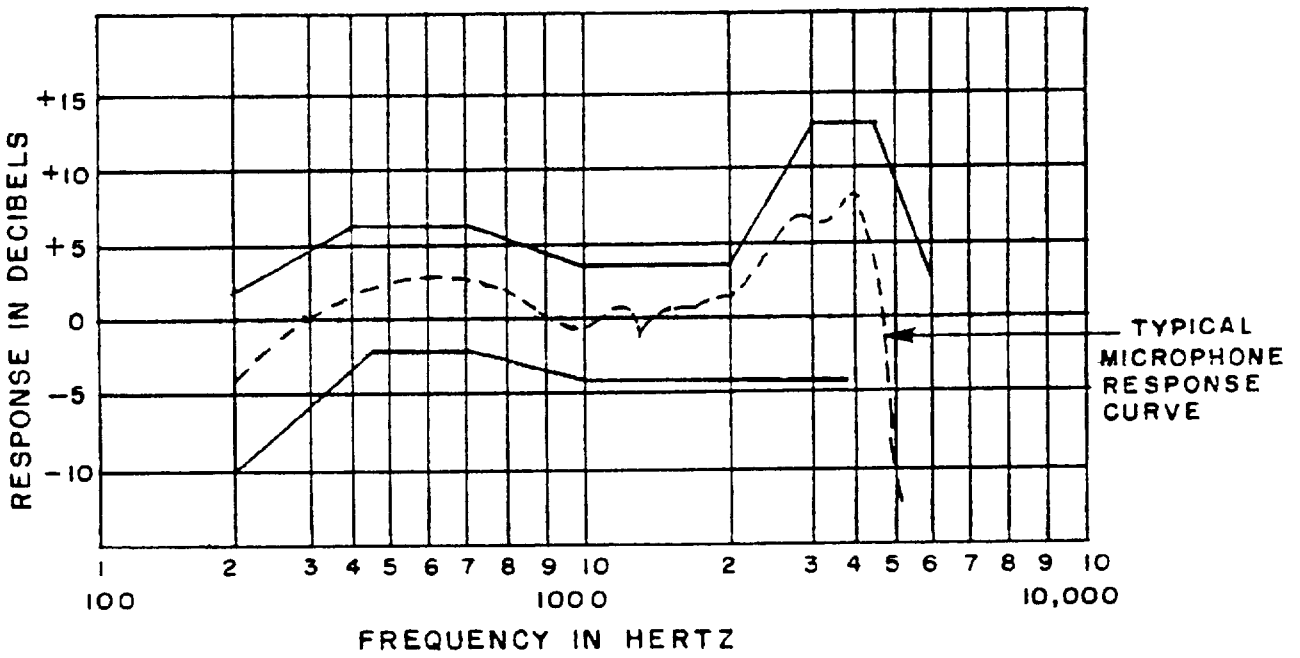
TOP VIEW OF BALL AND SOCKET JOINT

INCHES	MM	INCHES	MM
.001	0.03	.516	13.11
.015	0.38	.625	15.88
.06	1.5	.688	16.37
.075	1.91	.700	17.78
.200	5.16	.719	18.26
.250	6.35	.750	19.05
.297	7.54	1.000	25.40
.375	9.52	2.375	60.32
.490	12.37	3.500	88.90
		4.400	111.76

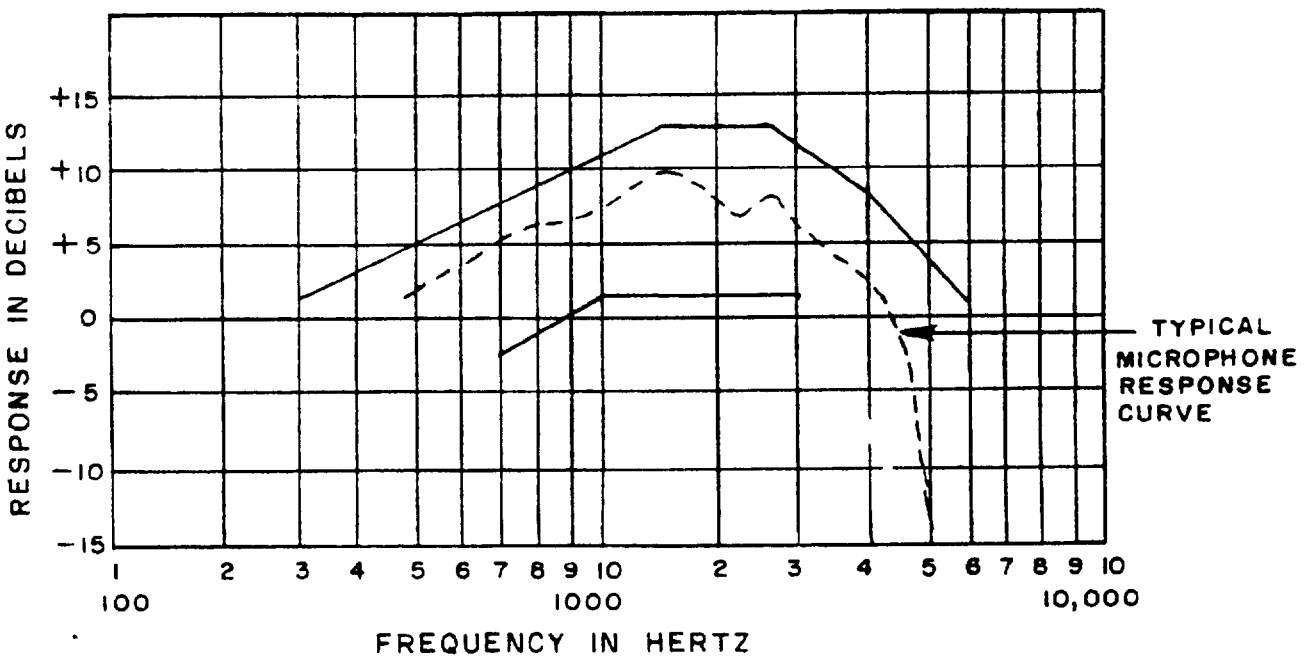
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

FIG. 5-1. Microprocessor assembly - Continued.

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Ⓒ FIGURE 2. Frequency response at sea level.



Ⓒ FIGURE 3. Frequency response at 25,000 feet.

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REQUIREMENTS

Color Semigloss black, 27038 in accordance with FED-STD-595.

Component parts

Microphone element See figure 1

Boom See figure 1.

Cord assembly: Electrical CX-4474/U

Plug assembly. U-173/U.

Weight: Not to exceed 45 grams.

Performance

Sensitivity at ground level 11 to 22 microvolts referred to 74 dB relative to 20 micropascals, with the microphone sound part 0.18 inch (4.7 mm) from and coaxial with the opening of the artificial voice

Sensitivity at altitude Sensitivity shall be within 3 dB of ground level sensitivity when tested at 25,000 feet.

Frequency response envelope shall be as shown on figures 2 and 3.

Impedance: 150 ohms ±15 ohms.

Total harmonic distortion Between 200-6,000 Hz, 5 percent at 105 dB relative to 20 micropascals.

Resistive load 150 ohms

Intended use The microphone, including the boom and wire assembly, is intended for use in new design headset-microphones

The microphone assembly shall be tested in accordance with the tests listed in table I

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© TABLE I. Parameter applicability

Inspection	Requirements paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1,3.4,3.5	4.7.1	X	X		
Sensitivity of ground level	3.5.1	4.7.2.2	X	X		
Sensitivity at altitude	3.5.2	4 7.2.3	X			
Frequency response at ground level	3.5.3	4.7.2.4	X	X		
Frequency response at altitude	3 5.4	4.7 2.5	X			
Impedance	3.5.5	4.7.3	X	X		
Noise cancellation characteristics	3 5.7.1	3.7.5.1	NA			
Effect of external magnetic field	3.5.8	4.7.6	X			
Stray magnetic field	3.5.9	4.7.7	X			
Linearity	3.5.10	4.7.8	X			
Talk out	3.5.11	4.7.9	X	X		
Dielectric withstanding voltage	3.5.12	4 7.10	X			
Signal to noise	3.5.7	4.7.5	X		X	
Distortion	3.5.6	4.7.4	X		X	
Interchangeability	3.5.13	4.7.11	X		X	
Group II						
Thermal shock	3.5.14	4.7.12	X			X
Humidity	3.5.15	4 7.13	X			X
Drop	3 5.16	4.7.14	X			X
Pressure equalization	3.5.17	4 7.15	X			X
Explosive decompression	3 5 18	4 7.16	X			X
Salt fog	3.5.19	4.7.17	X			X
Group III						
Vibration	3.5.20	4.7.18	X			X
Bounce	3.5.21	4.7.19	X			X
Altitude	3 5.22	4.7.20	X			X
Moisture barrier	3.5.23	4.7.21	X			X
Immersion	3.5.24	4.7.22	NA			
Group IV						
Fungus	3.5.25	4 7.23	NA			
Group V						
Gun blast	3.5.26	4 7 24	NA			

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Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES

User activities
Army - AR, AT, AV, ME
Navy - OS

Agent:
DLA - ES

Preparing activity
Air Force - 85

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